2. The Classics

Isaac and the little moon

If in the previous chapter I appeared to be saying that Plato and Aristotle have only done harm to the development of science, I would like to correct this impression. Aristotle's studies of nature – of botany and zoology, for example – are extraordinary scientific works, grounded upon meticulous observations of the natural world. The conceptual clarity, the attention to the variety of nature, the impressive intelligence and the openness of mind of the great philosopher made him an authority for centuries to come. The first systematic physics that we know of is Aristotle's, and it is not bad physics at all.

Aristotle presents it in a book entitled, precisely, *Physics*. The book didn't take its title from the name of a discipline: it was the discipline which got its name from Aristotle's book. For Aristotle, physics works as follows. First, it is necessary to distinguish between the heavens and Earth. In the heavens, everything is made up of a crystalline substance which moves in a circular motion and turns eternally around the Earth in great concentric circles, with the spherical Earth at the centre of everything. On the Earth, it is necessary to distinguish between forced motion and natural motion. Forced motion is caused by a thrust and ends when the thrust ends. Natural motion is vertical – upwards or downwards – and depends both on the substance and the location. Each substance has a 'natural place', that is to say, a proper altitude to which it always returns: earth at the bottom, water a little way above it, air a little higher still, and fire even higher. When you pick up a stone and let it fall, the stone moves downwards because it wants to return to its natural level. Air bubbles in water, fire in the air; and children's balloons move upwards, seeking their natural place.

Do not smile at this theory, or dismiss it, because it is very sound physics. It's a good and correct description of the motion of bodies immersed in a fluid and subject to gravity and friction, namely, the real things we meet in our everyday experience. It's not wrong physics, as is frequently said.[fn3] It's an approximation. But the physics of Newton, too, is an approximation of general relativity. And probably everything that we know today as well is an approximation of something else which we don't yet know. The physics of Aristotle is still rough, it is not quantitative (we cannot compute with it), but it